

Lesson 04

4.2 Well Ordering Principle

Before defining the Well Ordering Principle let's define the least element of a subset of real numbers, which is useful for defining Well Ordering Principle.

4.2.1 Definition-Least Element

Let A be a non empty subset of the set of real numbers. The real number l is called the least element of A if, $l \in A$ and for any $x \in A$, $x \geq l$.

According to the above definition you may realize that, there are some subsets of real numbers which doesn't have least elements.

4.2.2 Example:

Consider the following sub sets of the set of real numbers. Which one has a least element? Prove your answer.

- (i) $\mathbb{Z}$
- (ii) $[0, 1)$
- (iii) $(0, 1]$
- (iv) $\mathbb{N}$

Answer:

(i) It is clear that for the set $\mathbb{Z}$, it doesn't have a least element. Assume it exists. Let it be l . Since $l \in \mathbb{Z}$, $l - 1$ is also in $\mathbb{Z}$. But $l - 1 < l$, which contradict with the assumption.

(ii) The least element of $[0, 1)$ exists and is equal to 0. It is clear that $0 \in [0, 1)$ and for any $x \in [0, 1)$, $0 \leq x$.

(iii) The set $(0,1]$ doesn't have a least element. To prove this, assume it does exist, let it be l . Then $l \in (0,1]$. Hence $l > 0$. Now consider the real number $\frac{l}{2}$. Now it is clear that $0 < \frac{l}{2} < l$.

Hence $\frac{l}{2}$ is smaller than l and $\frac{l}{2} \in (0,1]$. This contradicts with the assumption.

(iv) The set $\mathbb{N}$ has a least element and is equal to 1. It is clear that $1 \in \mathbb{N}$ and for any $x \in \mathbb{N}$, $x \geq 1 \therefore 1$ is the least element of $\mathbb{N}$.

Activity. 1

Consider the following sub sets of the set of real numbers. Which one has a least element? Prove your answer.

(i) $\{n: n \in \mathbb{N}, n > 2017\}$

(ii) $\{n: n \in \mathbb{N}, n^2 - n + 1 \geq 0\}$

(iii) $\{q: q \in \mathbb{Q}, q > 2017\}$

(iv) $\{\frac{1}{n}: n \in \mathbb{N}\}$

(v) $\{r: r \text{ is irrational and } r \geq 2\}$

(vi) $\{\frac{1}{n}: n \in \mathbb{Z} \setminus \{0\}\}$.

From above examples and self-assessment questions, you can see that there are some sub sets of real numbers which do not have least elements. But if you choose any non-empty sub set of natural numbers, then it must have a least element. This principle is called Well Ordering Principle.

4.2.3 Well-Ordering Principle

Every non-empty subset of non-negative integers has a least element.

This principle plays a critical role in the proofs of this session and in subsequent sessions. Let us use it to show another property in the set of Natural Numbers known as the Archimedean Property.

4.3 Archimedean Property

4.3.1 Theorem

For any $a, b \in \mathbb{N}$, there exists a positive integer n such that $na \geq b$.

Proof:

We use the contradiction method to prove the Archimedean Property.

Assume that the Archimedean property is not true.

Then there exist, some positive integers a and b such that $na < b$ for every $n \in \mathbb{N}$.

Let $S = \{b - na | n \in \mathbb{N}\}$.

Since $na < b$ for every $n \in \mathbb{N}$, $b - na > 0$ for every $n \in \mathbb{N}$.

But $b - na$ is an integer for any $n \in \mathbb{N}$ and hence S is a non empty subset of the natural number set.

Hence by Well Ordering Principle, S must have a least element. Let it be $b - ma$, where $m \in \mathbb{N}$.

Now since $m \in \mathbb{N}$, $m + 1 \in \mathbb{N}$. $\therefore b - (m + 1)a \in S$.

But $b - (m + 1)a = (b - ma) - a < b - ma$.

This shows $b - (m + 1)a \in S$ and $b - (m + 1)a < b - ma$, which is contradict with $b - ma$ is

The least element of S .

Hence the assumption is false. $\therefore$ The Archimedean Property is true. ■

4.3.1 Examples

(i) Let $A = \{x + y + z + v : x, y, z, v, w \in \mathbb{N}, x^2 + y^2 + z^2 + v^2 = w^3\}$. Prove that the set A has a least element.

(ii) Let $B = \{m : m \in \mathbb{N} \text{ and } m^2 \text{ is divisible by } 2, 3, 4, 5, 6, 7, 8 \text{ and } 9\}$. Prove that the set B has a least element.

For each of these questions, there are two ways to prove the relevant sets have least elements. One way is concretely find the least element. In this way, using some techniques, you must find The least element and need to prove that is the least element. But for these two sets it is not trivial to find the least elements. Nonetheless if you succeed to guessing the least element correctly, you need to prove that is the least element.

The other method is direct use of the Well Ordering Principle. If each of these sets is a nonempty sub sets of the natural number set, then by Well Ordering Principle least element exists. In this way, we do not concretely find the least element. We prove, merely the existence of the least element. In following answers, we use the second method.

Answer: -

- (i) To apply well ordering principle, we must ensure two things, the set is non empty and the set is a sub set of a natural number set. Since , $y, z, v \in \mathbb{N}$, it is clear that $x + y + z + v$ also a member of $\mathbb{N}$. Hence the set A is a subset of $\mathbb{N}$. To prove the set A is non empty, you need some intelligent guess. Observe that $16+16+16+6 = 64$, this can be read as $4^2+4^2+4^2+4^2 = 4^3$. Then by taking $x = 4, y = 4, z = 4, v = 4$ and $= 4$, it is clear that $x + y + z + v = 16$ is an element of A . Hence A is a nonempty subset of $\mathbb{N}$. Therefore, by Well Ordering Principle the set A has a least element.
- (ii) It is trivial that B is a sub set of $\mathbb{N}$. Again, if we choose $m = 2.3.4.5.6.7.8.9$, then clearly m^2 is divisible by 2, 3, 4, 5, 6, 7, 8 and 9. Therefore 2.3.4.5.6.7.8.9 is an element of B and hence B is nonempty. Therefore, by Well Ordering Principle the set B has a least element.

4.3.2 Example

Find whether the least element exist for each of the following sets. Justify your answer.

- (i) $A = \{x + y : x, y \in \mathbb{N}, 13x - 7y = 1\}$,
(ii) $B = \{x - y : x, y \in \mathbb{N}, 13x - 7y = 1\}$,
(iii) $C = \{x + y : x, y \in \mathbb{N}, 9x - 6y = 5\}$

Answer:-

(i)

Clearly, since $x, y \in \mathbb{N}$, $x + y \in \mathbb{N}$. Hence A is a sub set of $\mathbb{N}$. Observe that $13 \times 6 - 7 \times 11 = 1$. Hence $6 + 11 = 17$ is a member of the set A . This shows that, the set A is a non empty subset of $\mathbb{N}$. Therefore by Well Ordering Principle the set A has a least element.

(ii)

It is clear that, some negative integers are members of B . $\{13 \times 6 - 7 \times 11 = 1$. Hence $6 - 11 = -5$ is a member of the set $B\}$. We know that, for some sub sets of $\mathbb{Z}$ have least elements and for some none empty sub sets of $\mathbb{Z}$ doesn't have.

Assume that the set B has a least element. Let it be L . Hence there exist $a, b \in \mathbb{N}$ such that $L = a - b$ and $13a - 7b = 1$. Now let $x = a + 7$ and $y = b + 13$. Then clearly $x, y \in \mathbb{N}$ and $13x - 7y = 13(a + 7) - 7(b + 13) = 13a - 7b = 1$. Hence $x - y \in B$. Also observe that $x - y = (a + 7) - (b + 13) = a - b - 6 = L - 6 < L$. Which contradict with L is the east element of B . Hence our assumption is wrong and therefore the set B doesn't have a least element.

(iii)

Again it is clear that the set C is a sub set of $\mathbb{N}$. If we can prove C is non empty, then as in (i), we can apply the Well Ordering Principle to prove the existence of the least element.

But observe that, if there are two positive integers x and y such that $9x - 6y = 5$ then

$3(3x - 2y) = 5$. Hence $(3x - 2y) = \frac{5}{3}$ this cannot happen, since $3x - 2y$ is an integer.

Hence the set C is empty. Hence C doesn't have a least element.

Activity. 2

- (a) Does the set $\{113n - 355 : n \in \mathbb{N}\}$ has a least element? Justify your answer.
- (b) Does the set $\{355 - 113n : n \in \mathbb{N}\}$ has a least element? Justify your answer.
- (c) Guess the least element of the set $\{x + y + z : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\}$. Prove your guessing is correct.
- (d) According to the Well Ordering Principle, does the least element of the set $\{x : x \in \mathbb{N}, x^2 + 1 \text{ is a multiple of } 3\}$ exist? Justify your answer.

Activity. 3

Consider the following sets and complete the activity given below.

$$A = \{x + y - z : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\},$$

$$B = \{z - x - y : x, y, z \in \mathbb{N}, x^2 + y^2 = z^2\},$$

- (i) Show that $A \neq \emptyset$.
- (ii) Using a well known property in a triangle, show that any element of A is positive.
- (iii) Using algebraic methods, show that any element of A is positive.
- (iv) Show that $B \neq \emptyset$.
- (v) Prove that B doesn't have a least element. {Hint: if $5 - 4 - 3 \in B$ then does the element $5k - 4k - 3k \in B$?

?

Does the set $\{x + y : x, y \in \mathbb{N}, x, y \text{ both odd and } x^2 + y^2 \text{ is a perfect squarer}\}$ has a least element?

4.4 Principle of Mathematical Induction

Here we prove the Principle of Mathematical Induction using the Well Ordering Principle. Notice that for the convenience of applying the Well Ordering Principle, the appearance of the following theorem is slightly different from the familiar form.

4.4.1 Theorem - Principle of Mathematical Induction

Let S be a subset of positive integers (i.e. $S \subseteq \mathbb{N}$) with the following properties:

- (i) The integer 1 belongs to S . (i.e. $1 \in S$)
- (ii) Whenever the integer k in S , the next integer $k + 1$ must also be in S .

Then the set S is equal to the set of natural numbers.

Proof:

Let $T = \mathbb{N} \setminus S$. i.e. the set T consists of all positive integers which are not in S .

(Recall that if A and B are two sets then the set $A \setminus B = \{x \mid x \in A, x \notin B\}$.)

Assume that $T \neq \phi$. $\therefore T$ is a non empty sub set of natural numbers.

Hence by well ordering principle T has a least element. Let it be a .

From property (i), $1 \in S$.

$\therefore 1 \notin T$.

$\therefore a \neq 1$.

$\therefore a > 1$.

Now since a is the least element of T , $a - 1 \notin T$.

$\therefore a - 1 \in S$.

Now by property (ii), $(a - 1) + 1 = a \in S$. This contradicts with the fact $a \in T$.

Hence the assumption is false.

$\therefore T = \phi$

$\therefore S = \mathbb{N}$. ■

4.4.2 Example:

Use the Principle of Mathematical Induction to show that, for each $n \in \mathbb{N}$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \dots\dots\dots(1)$$

Proof:

Let $S = \left\{ n \mid n \in \mathbb{N}, 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \right\}$. i.e. S be the set of positive integers which satisfies the equation (1).

Observe that $1 = \frac{1(1+1)}{2}$ i.e. the equation (1) is true for $n = 1$.

$\therefore 1 \in S$(α)

Let $k \in S$ (Where k is fixed but unspecified integer).

Then $1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$.

If we add $k + 1$ to the both sides, then $1 + 2 + 3 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1)$.

But $\frac{k(k+1)}{2} + (k + 1) = \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}$.

Hence for $n = k + 1$ the equation (1) is true. This implies $(k + 1) \in S$.

$\therefore$ Whenever $k \in S$, the integer $k + 1 \in S$(β)

By (α), (β) the set S satisfies the two properties of the Principle of Mathematical Induction.

$\therefore S = \mathbb{N}$.

Hence for each $n \in \mathbb{N}$, $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. ■

Remark

When giving induction proofs, the most common method is shorten the argument by eliminating all reference to the set S , and proceed to show that the result in the problem(or its modification) is true for the integer 1, and if it is true for the integer k , then it is also true for $k + 1$.

Theorem (The Principle of Mathematical Induction)

Let $P(n)$ be any statement about n .

Assume that (i) $P(1)$ is true

and (ii) for each natural number n , if $P(n)$ is true then $P(n + 1)$ is true.

Then for each natural number n , $P(n)$ is true.

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Theorem (Strong Form of Principle of Mathematical Induction)

Let $P(n)$ be a statement about n .

Assume that (i) $P(1)$ is true,

(ii) for each natural number n , if $P(1), P(2), \dots, P(n)$ are all true, then $P(n + 1)$ is true.

Then for each natural number n , $P(n)$ is true.

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Though the two versions are equivalent, from the following activities and from the review questions appeared at the end of this session, you may realize the importance of both versions of the Principle of Mathematical Induction.

Activity 5

Let $s_0 = a$ and $s_n = 2s_{n-1} + b$ for $n \in \mathbb{N}$, where a and b are real constants.

- (a) Calculate the first five terms of s_n .
- (b) By observing these, can you guess a formula for s_n ? {depend on a, b and n }
- (c) Use suitable version of Principle of Mathematical Induction to prove your guess.

Activity 6

Let $f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$ for $n \in \mathbb{N}$. Using a suitable version of mathematical induction, prove that f_n is an integer for each $n \in \mathbb{N}$.
